SignRR: Retrieve and Refine Real Motion for Sign Language Production

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Motivation

Task. Sign language production turns a sequence of glosses, the written labels of signs, into continuous 3D signing.

Generative models synthesize every frame. The motion is smooth, but rare handshapes and each signer's own style tend to be averaged out.

Retrieval methods reuse real recorded segments. The hands look natural, but pieces from different signers and speeds break the rhythm of the sentence.

GENERATIVE	RETRIEVAL	SignRR (OURS)
Smooth motion, but the hands lose detail.	Real hands, but the rhythm breaks.	Real motion, refined into coherent signing.

Retrieve-and-Refine Pipeline

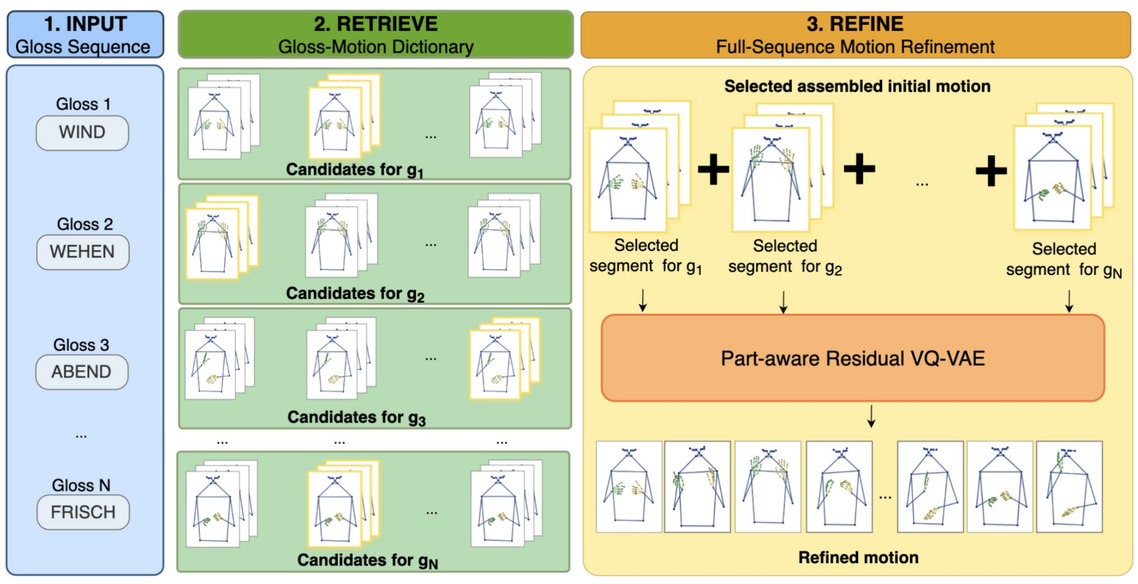


Figure 1: SignRR pipeline: per-gloss retrieval, assembly, and full-sequence refinement.

Method

Dictionary. Built automatically from continuous training videos. At test time, dynamic programming picks one segment per gloss so that neighbours join smoothly.

Part-aware RVQ. The body and each hand have their own encoder and residual codebooks, so small finger movements are not lost.

Length alignment. A gloss encoder predicts how long the sentence should be, and a single decoder produces the final motion.

Training. We train on random assemblies of dictionary segments, so the model learns to turn them into real, continuous signing.

$$\hat{\mathbf{z}}_l^p = \mathrm{VQ}\big(r_{l-1}^p; C_l^p\big)\,, \qquad r_l^p = r_{l-1}^p - \mathrm{sg}\big[\hat{\mathbf{z}}_l^p\big]\,, \qquad l = 1, \ldots, L\,,$$

Residual quantization: each layer quantizes the residual of the previous layer.

Conclusion

Retrieve, then refine. Real segments give natural articulation, and refining the whole sentence makes it coherent.

Results. SignRR gets the best back-translation scores on PHOENIX14T and CSL-Daily, with a competitive pose error, and it runs about 25× faster than G2P-DDM.

Next. We plan to add facial (non-manual) features and to drive photorealistic signing avatars.

Experimental Setup

Data. We evaluate on PHOENIX14T (German) and CSL-Daily (Chinese), using 3D poses with 61 keypoints for the body and both hands.

Evaluation. A Sign Language Transformer trained on real poses translates our poses back into text (WER, BLEU, ROUGE). We also report the joint position error (MPJPE).

Results

Method	DEV					TEST				
	WER↓	BLEU-1↑	BLEU-4↑	ROUGE↑	MPJPE↓	WER↓	BLEU-1↑	BLEU-4↑	ROUGE↑	MPJPE↓
GT	64.67	30.88	13.38	31.11	0.00	63.61	31.30	13.43	31.03	0.00
Sign-IDD† [25]	86.34	26.49	10.79	27.34	**40.74**	85.98	25.34	9.81	24.91	**50.19**
G2P-DDM† [31]	85.43	26.97	11.05	28.13	44.41	83.92	27.04	10.78	27.56	54.28
Miao et al.† [15]	95.62	15.97	4.60	16.94	45.57	95.66	14.16	3.29	14.61	56.17
Sign Stitching [29]	–	–	–	–	–	–	25.14	6.67	26.49	–
PT base [17]	–	–	–	–	–	–	6.27	1.59	9.50	–
PT + GN [17]	–	–	–	–	–	–	11.45	4.04	14.52	–
Ours	**82.07**	**28.16**	**11.72**	**28.88**	43.83	**80.98**	**28.78**	**10.96**	**28.15**	53.70

Table 1: Results on PHOENIX14T. Bold: best; underlined: second best; †: reproduced under our setting.

Method	DEV				TEST			
	BLEU-1↑	BLEU-4↑	ROUGE↑	MPJPE↓	BLEU-1↑	BLEU-4↑	ROUGE↑	MPJPE↓
GT	23.59	5.87	23.51	0.00	23.12	5.06	23.06	0.00
Sign-IDD† [25]	11.66	0.00	11.83	135.71	11.98	0.00	11.98	147.51
Miao et al.† [15]	11.43	0.23	11.30	**91.29**	11.53	0.20	11.46	**99.70**
Ours	**15.23**	**1.34**	**15.01**	106.21	**15.45**	**1.57**	**15.46**	115.40

Table 2: Results on CSL-Daily. Bold: best; underlined: second best; †: reproduced under our setting.

Analysis

Variant	Configuration		Back-translation				
	Selection	Refine.	WER↓	BLEU-1↑	BLEU-4↑	CHRF↑	ROUGE↑
Retrieve only	DP	✗	83.78	22.06	8.31	27.86	21.47
Retrieve only	Random	✗	81.50±0.65	23.94±0.74	9.13±0.40	29.36±0.22	23.27±0.73
SignRR (Random sel.)	Random	✓	81.11±0.21	28.23±0.06	10.76±0.17	32.61±0.14	27.64±0.25
SignRR (Proposed)	DP	✓	**80.98**	**28.78**	**10.96**	**32.90**	**28.15**

Table 3: Selection vs. refinement, PHOENIX14T test. Random: mean ± std over 3 seeds.

Variant	Configuration		Back-translation				
	Training	Cross-attn	WER↓	BLEU-1↑	BLEU-4↑	CHRF↑	ROUGE↑
Synthetic input	Synthetic	✗	99.04	10.39	1.01	19.70	8.11
Synthetic input	Synthetic	✓	90.02	25.06	8.72	30.13	23.99
SignRR (Proposed)	Dictionary	✗	**80.98**	**28.78**	**10.96**	**32.90**	**28.15**
SignRR + gloss	Dictionary	✓	83.47	27.03	10.24	31.84	27.17

Table 4: Refiner trained on synthetic motion vs. real segments, PHOENIX14T test.

Refinement brings the gain, and it needs real motion. With the same retrieved segments, refinement raises BLEU-1 from 22.06 to 28.78. Trained on synthetic motion instead, the refiner reaches only 10.39, or 25.06 with gloss attention.

Refinement Model: Part-aware Residual VQ-VAE

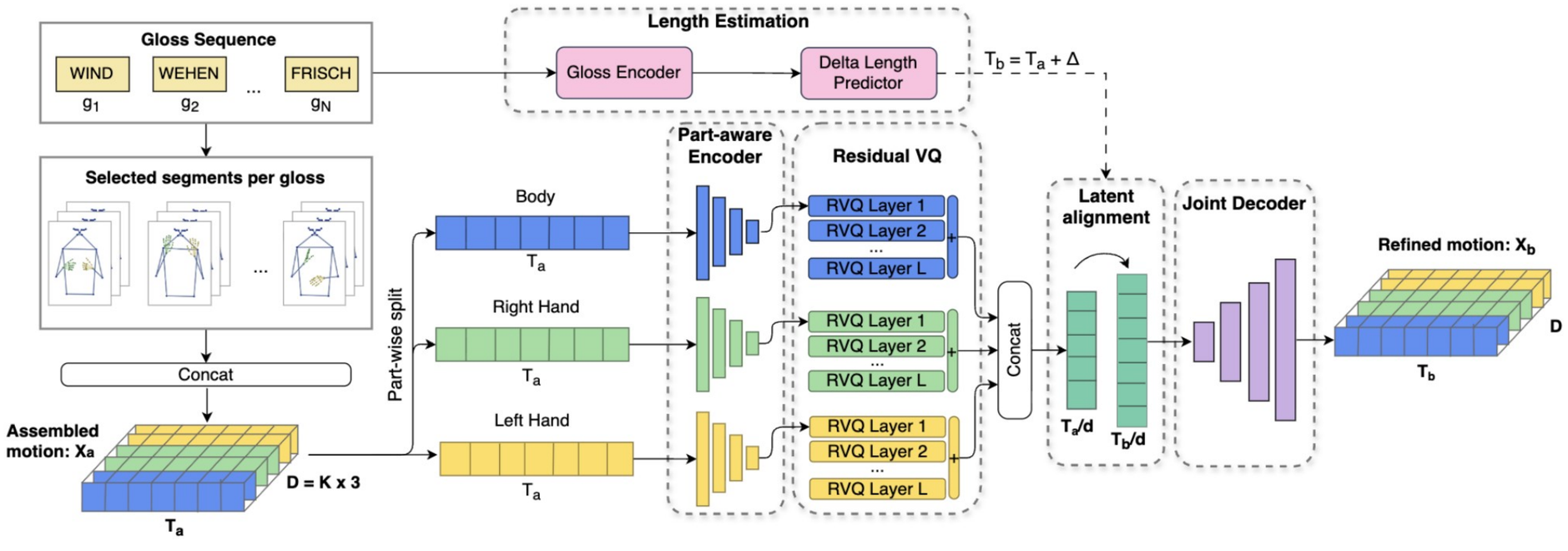


Figure 2: Part-aware Residual VQ-VAE: separate body and hand encoders with residual VQ, latent length alignment, and a joint decoder.